The Definition of the Derivative

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Difference Quotients & Limit Definitions**

Let $f(x)$ be a function such that $f'(x) = \cos(x^2)$ and $g(x)$ be a function defined by $g(x) = \sqrt{x}$. ; ; \textbf{(a)} Find the derivative of the composite function $h(x) = f(g(x))$. ; ; \textbf{(b)} If $f(1) = \pi$, find the value of $h'(4)$.

\textbf{(a)} We use the chain rule to find the derivative of $h(x) = f(g(x))$. ; The chain rule states that $h'(x) = f'(g(x)) \cdot g'(x)$. ; We are given $f'(x) = \cos(x^2)$ and $g(x) = \sqrt{x}$. ; First, we find $g'(x)$: $g'(x) = \frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$. ; Next, we find $f'(g(x))$ by substituting $g(x)$ into $f'(x)$: $f'(g(x)) = \cos((\sqrt{x})^2) = \cos(x)$. ; Therefore, $h'(x) = f'(g(x)) \cdot g'(x) = \cos(x) \cdot \frac{1}{2\sqrt{x}} = \frac{\cos(x)}{2\sqrt{x}}$. ; ; \textbf{(b)} To find $h'(4)$, we substitute $x = 4$ into our expression for $h'(x)$: ; $h'(4) = \frac{\cos(4)}{2\sqrt{4}} = \frac{\cos(4)}{2 \cdot 2} = \frac{\cos(4)}{4}$. ; ; Note: The information $f(1) = \pi$ is extraneous for this specific part of the question.

2 **[Foundational] • Target: Differentiability Implies Continuity**

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.

$f'(x) = 3(3)x^2 - 4 = 0 \Rightarrow x = \pm\frac{2}{3}$. Since $f''(x) = 18x$, $f''(-\frac{2}{3}) < 0$ (local maximum) and $f''(\frac{2}{3}) > 0$ (local minimum). Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

3 **[Foundational] • Target: Tangent and Normal Lines**

Let $f(x) = \sin(x^2)$. What is the derivative of $f(x)$?

To find the derivative of $f(x) = \sin(x^2)$, we use the chain rule. The chain rule states that if $f(x) = g(h(x))$, then $f'(x) = g'(h(x)) \cdot h'(x)$. In this case, let $g(u) = \sin(u)$ and $h(x) = x^2$. Then $g'(u) = \cos(u)$ and $h'(x) = 2x$. Applying the chain rule, we get $f'(x) = \cos(x^2) \cdot 2x$, which simplifies to $2x\cos(x^2)$.

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4 [Foundational] • Target: Estimating Derivatives from Tables & Graphs

The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.

The function is given by $f(x) = ax - 3$. The area of the shaded region is represented by the definite integral of $f(x)$ from $x = 0$ to $x = 4$. We are given that this area is 20. $\int_0^4 (ax - 3)\,dx = 20$ First, find the antiderivative of $ax - 3$: $\int (ax - 3)\,dx = \frac{a}{2}x^2 - 3x + C$ Now, evaluate the definite integral from 0 to 4: $\left[\frac{a}{2}x^2 - 3x\right]_0^4 = \left(\frac{a}{2}(4)^2 - 3(4)\right) - \left(\frac{a}{2}(0)^2 - 3(0)\right) = \left(\frac{a}{2}(16) - 12\right) - (0) = 8a - 12$ We are given that this area is 20: $8a - 12 = 20$ Now, solve for a: $8a = 20 + 12$ $8a = 32$ $a = \frac{32}{8}$ $a = 4$ Therefore, the value of the parameter a is 4.

5 [Foundational] • Target: Difference Quotients & Limit Definitions

Consider the function $f(x) = x^2 - 3x$. If $g(x)$ is a function such that $h(x) = f(g(x))$, and at $x = 2$, $g(2) = 5$ and $g'(2) = 3$, find $h'(2)$.

This question involves the chain rule for differentiation. The chain rule states that if $h(x) = f(g(x))$, then $h'(x) = f'(g(x)) \cdot g'(x)$. First, we find the derivative of $f(x)$: $f'(x) = \frac{d}{dx}(x^2 - 3x) = 2x - 3$. Now, we need to find $f'(g(x))$. Since $g(x)$ is a general function, we substitute $g(x)$ into $f'(x)$: $f'(g(x)) = 2g(x) - 3$. To find $h'(2)$, we apply the chain rule formula at $x = 2$: $h'(2) = f'(g(2)) \cdot g'(2)$. We are given that $g(2) = 5$ and $g'(2) = 3$. Substitute $g(2) = 5$ into $f'(g(x))$: $f'(g(2)) = 2(5) - 3 = 10 - 3 = 7$. Now, multiply $f'(g(2))$ by $g'(2)$: $h'(2) = 7 \cdot 3 = 21$. Therefore, the value of $h'(2)$ is 21.

6 [Foundational] • Target: Differentiability Implies Continuity

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?

The velocity of the particle is given by $v(t) = t^2 - 6t + 8$. Acceleration is the rate of change of velocity with respect to time, which means acceleration is the derivative of the velocity function, $a(t) = v'(t)$. Differentiating $v(t)$ with respect to t, we get: $a(t) = \frac{d}{dt}(t^2 - 6t + 8) = 2t - 6$. We need to find the instantaneous acceleration at time $t = 3$ seconds. Substituting $t = 3$ into the acceleration function: $a(3) = 2(3) - 6 = 6 - 6 = 0$ m/s^2. Therefore, the instantaneous acceleration of the particle at time $t = 3$ seconds is 0 m/s^2.

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7 [Foundational] • Target: Tangent and Normal Lines

A particle moves along a straight line so that its velocity at time $t \geq 0$ is modeled by $v(t) = 6t^2 - 18t + 12$. At what time t is the acceleration of the particle equal to 0?

To find the acceleration $a(t)$, take the derivative of the velocity function $v(t)$ with respect to time t: $a(t) = v'(t) = \frac{d}{dt}(6t^2 - 18t + 12) = 12t - 18$. Set $a(t) = 0$ and solve for t: $12t - 18 = 0 \Rightarrow 12t = 18 \Rightarrow t = \frac{18}{12} = \frac{3}{2}$. Common Misconceptions: - Solving $v(t) = 0$ yields $6(t-1)(t-2) = 0 \Rightarrow t = 1$ or $t = 2$. These are the times when the particle is at rest, not when its acceleration is zero. - Omitting the factor of 2 when applying the power rule gives $6t - 18 = 0 \Rightarrow t = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Consider the function $f(x)$ whose graph is shown above. The graph of $f(x)$ consists of line segments and a quarter circle. Let $g(x) = \int_0^x f(t)\,dt$.

By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. We need to find $g'(2)$, which means we need to evaluate $f(2)$. From the graph, at $x = 2$, the function $f(x)$ is on a line segment connecting $(0, 4)$ and $(4, 0)$. The equation of this line segment can be found. The slope is $m = \frac{0-4}{4-0} = -1$. Using the point-slope form with $(4, 0)$, we get $y - 0 = -1(x - 4)$, which simplifies to $y = -x + 4$. Therefore, $f(x) = -x + 4$ for $0 \leq x \leq 4$. Evaluating $f(2)$ gives $f(2) = -2 + 4 = 2$. Thus, $g'(2) = f(2) = 2$.

9 [Intermediate] • Target: Difference Quotients & Limit Definitions

What is the value of $\lim_{h \to 0} \frac{(2+h)^4 - 16}{h}$?

By definition, the derivative of a function f at $x = a$ is given by: $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ The expression $\lim_{h \to 0} \frac{(2+h)^4 - 16}{h}$ matches this definition for the function $f(x) = x^4$ evaluated at $a = 2$, since $f(2) = 2^4 = 16$. Applying the power rule to find the derivative: $f'(x) = 4x^3$ Evaluating at $x = 2$: $f'(2) = 4(2)^3 = 4(8) = 32$ Common Misconceptions:

10 [Intermediate] • Target: Differentiability Implies Continuity

Use L'Hopital's Rule to evaluate $\lim_{x \to \infty} \frac{7x}{e^{4x}}$.

This is $\frac{\infty}{\infty}$ form. Applying L'Hopital's Rule: $\lim_{x \to \infty} \frac{7}{4e^{4x}} = 0$, since the exponential grows without bound. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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11 **[Intermediate]** • **Target: Tangent and Normal Lines**

A 5-ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. Find the rate at which the top of the ladder is sliding down the wall when the bottom is 3 ft from the wall.

$x^2 + y^2 = 5^2$. At $x = 3$, $y = 4$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{3}{2}$ ft/s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

12 **[Intermediate]** • **Target: Estimating Derivatives from Tables & Graphs**

Find $\frac{d}{dx}[\sin((-2x-2)^3)]$.

Chain rule (twice): $\cos((-2x-2)^3) \cdot 3(-2x-2)^2 \cdot -2 = -6(-2x-2)^2\cos((-2x-2)^3)$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

13 **[Intermediate]** • **Target: Difference Quotients & Limit Definitions**

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

The displacement of the particle is the definite integral of its velocity function over the given time interval. \\ Displacement = $\int_0^4 v(t)\,dt = \int_0^4 5\,dt$. \\ To evaluate this definite integral, we find the antiderivative of USD 5, *whichis*5t*x*[5t]_{0}^{4} = 5(4) - 5(0) = 20 - 0 = 20. \\ Therefore, the total displacement of the particle is 20 units.

14 **[Intermediate]** • **Target: Differentiability Implies Continuity**

Find the area of the region bounded by $y = 28x$ and $y = x^2$ in the first quadrant.

Intersections at $x = 0$ and $x = 28$. Area $= \int_0^{28}(28x - x^2)\,dx = \left[\frac{28}{2}x^2 - \frac{x^3}{3}\right]_0^{28} = \frac{21952}{6}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Tangent and Normal Lines** • **Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

Find $\frac{d}{dx}[\sin((1-2x)^3)]$.

Chain rule (twice): $\cos((1-2x)^3) \cdot 3(1-2x)^2 \cdot -2 = -6(1-2x)^2\cos((1-2x)^3)$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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16 **[Challenging]** • **Target: Estimating Derivatives from Tables & Graphs** • **Cross-Lesson: 3.01 The Chain Rule**

Given $\int_{-2}^{-1} 1x^2\,dx = \frac{7}{3}$ and $\int_{-1}^{0} 1x^2\,dx = \frac{1}{3}$, find $\int_{-2}^{0} 1x^2\,dx$.

By additivity of the definite integral: $\int_{-2}^{0} = \int_{-2}^{-1} + \int_{-1}^{0} = \frac{7}{3} + \frac{1}{3} = \frac{8}{3}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

17 **[Challenging]** • **Target: Difference Quotients & Limit Definitions** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Given $\int_{-1}^{0} 4x^2\,dx = \frac{4}{3}$ and $\int_{0}^{1} 4x^2\,dx = \frac{4}{3}$, find $\int_{-1}^{1} 4x^2\,dx$.

By additivity of the definite integral: $\int_{-1}^{1} = \int_{-1}^{0} + \int_{0}^{1} = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

18 **[Challenging]** • **Target: Differentiability Implies Continuity** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, we find the derivative of $f(x) = x^2 - 6x + 10$. Using the power rule, $f'(x) = 2x - 6$. Next, we evaluate the derivative at $x = 4$: $f'(4) = 2(4) - 6 = 8 - 6 = 2$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 4$ is 2.

19 **[Challenging]** • **Target: Tangent and Normal Lines** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Given $\int_{-2}^{1} 5x^2\,dx = 15$ and $\int_{1}^{5} 5x^2\,dx = \frac{620}{3}$, find $\int_{-2}^{5} 5x^2\,dx$.

By additivity of the definite integral: $\int_{-2}^{5} = \int_{-2}^{1} + \int_{1}^{5} = 15 + \frac{620}{3} = \frac{665}{3}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

20 **[Challenging]** • **Target: Estimating Derivatives from Tables & Graphs** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

Given $\int_{0}^{3} 4x^2\,dx = 36$ and $\int_{3}^{5} 4x^2\,dx = \frac{392}{3}$, find $\int_{0}^{5} 4x^2\,dx$.

By additivity of the definite integral: $\int_{0}^{5} = \int_{0}^{3} + \int_{3}^{5} = 36 + \frac{392}{3} = \frac{500}{3}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Difference Quotients & Limit Definitions

Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ ✓

b. $\lim_{h\to 0} \frac{f(x+h)+f(x)}{h}$

c. $\lim_{h\to \infty} \frac{f(x+h)-f(x)}{h}$

d. $\frac{f(x+h)-f(x)}{h}$

By definition, $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$.

22 [Foundational] • Target: Difference Quotients & Limit Definitions

Evaluate $\lim_{h\to 0} \frac{(3+h)^2-9}{h}$.

a. 3

b. 6 ✓

c. 9

d. 0

This limit represents $f'(3)$ for $f(x) = x^2$. $f'(x) = 2x \Rightarrow f'(3) = 2(3) = 6$.

23 [Foundational] • Target: Difference Quotients & Limit Definitions

Which function and value of c are represented by $\lim_{x\to 4} \frac{\sqrt{x}-2}{x-4}$?

a. $f(x) = \sqrt{x}$ **at** $c = 4$ ✓

b. $f(x) = x^2$ at $c = 2$

c. $f(x) = \sqrt{x}$ at $c = 2$

d. $f(x) = \frac{1}{\sqrt{x}}$ at $c = 4$

Using the alternate form $f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c}$, this corresponds to $f(x) = \sqrt{x}$ at $c = 4$.

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24 [Foundational] • Target: Differentiability Implies Continuity

If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. **f is continuous at $x = c$** ✓

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

Differentiability implies continuity. (The converse is false, as demonstrated by sharp corners like $f(x) = |x|$).

25 [Foundational] • Target: Differentiability Implies Continuity

Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. **The left-hand and right-hand derivative limits are unequal (corner)** ✓

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

For $x < 3$, $f'(x) = -1$, and for $x > 3$, $f'(x) = 1$. The corner creates unequal one-sided derivative limits.

26 [Foundational] • Target: Differentiability Implies Continuity

The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. **A vertical tangent line where $f'(x) \to \infty$** ✓

b. A jump discontinuity

c. A sharp cusp

d. A hole

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. As $x \to 0$, $f'(x) \to \infty$, which corresponds to a vertical tangent line.

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27 [Foundational] • Target: Tangent and Normal Lines

What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$ ✓

b. $y - 4 = 8(x - 2)$

c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

$f'(x) = 3x^2 - 2$. Slope at $x = 2$ is $f'(2) = 3(4) - 2 = 10$. In point-slope form: $y - 4 = 10(x - 2)$.

28 [Foundational] • Target: Tangent and Normal Lines

What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?

a. $-\frac{1}{5}$ ✓

b. 5

c. -5

d. $\frac{1}{5}$

$y' = 2x + 3 \Rightarrow y'(1) = 5$. The normal line is perpendicular to the tangent line, so its slope is $-\frac{1}{5}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent and Normal Lines

At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$ ✓

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

Horizontal tangent requires $y' = 0$: $2x - 4 = 0 \Rightarrow x = 2$. $y(2) = 4 - 8 + 3 = -1$. Point is $(2, -1)$.

30 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10 ✓

b. 8

c. 16

d. 12

Symmetric difference quotient: $s'(4) \approx \frac{s(6) - s(2)}{6 - 2} = \frac{54 - 14}{4} = \frac{40}{4} = 10$.

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31 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0 ✓

b. 1

c. Positive

d. Undefined

By Fermat's theorem, the derivative at an interior local extremum of a differentiable function is zero: $f'(5) = 0$.

32 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f **is strictly increasing** ✓

b. f is strictly decreasing

c. f is concave up

d. f is linear

A strictly positive first derivative implies that the function is strictly increasing on the interval.

33 [Intermediate] • Target: Difference Quotients & Limit Definitions

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

a. **5** ✓

b. 12

c. -3

d. 9

The derivative of $f(x)$ is given as $f'(x) = 3x^2 - 12x + 9$. To find the function $f(x)$, we integrate $f'(x)$: $f(x) = \int (3x^2 - 12x + 9)\,dx = x^3 - 6x^2 + 9x + C$. We are given that $f(x)$ passes through $(0, 5)$. Substituting $x = 0$ and $f(0) = 5$ into the expression for $f(x)$: $5 = (0)^3 - 6(0)^2 + 9(0) + C$, which gives $C = 5$. Thus, the function is $f(x) = x^3 - 6x^2 + 9x + 5$. The derivative at $x = 0$ is $f'(0) = 3(0)^2 - 12(0) + 9 = 9$. The point on the graph at $x = 0$ is $(0, f(0)) = (0, 5)$. The equation of the tangent line at $x = 0$ has the form $y - y_1 = m(x - x_1)$, where $m = f'(0)$ and $(x_1, y_1) = (0, 5)$. So, the equation of the tangent line is $y - 5 = 9(x - 0)$, which simplifies to $y = 9x + 5$. The y-intercept is the value of y when $x = 0$, which is 5. Therefore, the y-intercept of the tangent line is 5.

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34 [Intermediate] • Target: Differentiability Implies Continuity

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The equation of a tangent line to a function $f(x)$ at a point $(a, f(a))$ is given by $y - f(a) = f'(a)(x - a)$. We are given that the tangent line at $x = 1$ has the equation $y = 2x + k$. The slope of this tangent line is 2, which means $f'(1) = 2$. We are also given that $f(1) = 3$. Substituting these values into the point-slope form of the tangent line equation: $y - 3 = 2(x - 1)$ $y - 3 = 2x - 2$ $y = 2x - 2 + 3$ $y = 2x + 1$ Comparing this to the given tangent line equation $y = 2x + k$, we can see that $k = 1$. Alternatively, since the point $(1, f(1))$ lies on the tangent line, we can substitute $x = 1$ and $y = f(1) = 3$ into the equation of the tangent line: $3 = 2(1) + k$ $3 = 2 + k$ $k = 3 - 2$ $k = 1$.

35 [Intermediate] • Target: Tangent and Normal Lines

Consider the function $f(x) = x^2 - 5x + 6$. What is the limit of $f(x)$ as x approaches 2?

a. **-1** ✓

b. 0

c. 1

d. 2

To find the limit of the polynomial function $f(x) = x^2 - 5x + 6$ as x approaches 2, we can directly substitute $x = 2$ into the function because polynomials are continuous everywhere.
$\lim_{x \to 2} (x^2 - 5x + 6) = (2)^2 - 5(2) + 6 = 4 - 10 + 6 = 0$ Therefore, the limit is 0.

36 [Intermediate] • Target: Estimating Derivatives from Tables & Graphs

Consider the function $f(x)$ graphed below. What is the limit of $f(x)$ as x approaches 1, i.e., $\lim_{x \to 1} f(x)$?

a. -1

b. **0** ✓

c. 1

d. 2

The limit of a function as x approaches a certain value is the y-value that the function approaches from both the left and the right. Observing the graph, as x gets closer and closer to 1 from both sides, the y -value of the function approaches 0.

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37 [Intermediate] • Target: Difference Quotients & Limit Definitions

A region R is bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi\ln(10)$

b. $2\pi\ln(5)$

c. $\pi\ln(5)$ ✓

d. $2\pi\ln(10)$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and $x = a, x = b$ about the y-axis can be found using the shell method. The formula for the volume is $V = \int_a^b 2\pi x f(x)\,dx$. \\ In this case, $f(x) = \frac{1}{x^2+1}$, $a = 1$, and $b = 3$. \\ So, the volume is $V = \int_1^3 2\pi x\left(\frac{1}{x^2+1}\right)dx$. \\ We can pull out the constant 2π: $V = 2\pi\int_1^3 \frac{x}{x^2+1}\,dx$. \\ To evaluate this integral, we use a substitution. Let $u = x^2 + 1$. Then, $du = 2x\,dx$, which means $x\,dx = \frac{1}{2}du$. \\ We also need to change the limits of integration: \\ When $x = 1$, $u = 1^2 + 1 = 2$. \\ When $x = 3$, $u = 3^2 + 1 = 10$. \\ Substituting into the integral: $V = 2\pi\int_2^{10} \frac{1}{u}\cdot\frac{1}{2}du$. \\ $V = \pi\int_2^{10} \frac{1}{u}\,du$. \\ The integral of $\frac{1}{u}$ with respect to u is $\ln|u|$. \\ So, $V = \pi[\ln|u|]_2^{10}$. \\ Evaluating at the limits: $V = \pi(\ln(10) - \ln(2))$. \\ Using the logarithm property $\ln(a) - \ln(b) = \ln(\frac{a}{b})$: $V = \pi\ln(\frac{10}{2})$. \\ $V = \pi\ln(5)$. \\ Therefore, the volume of the solid is $\pi\ln(5)$. \\ Common misconception: Students might try to use the disk or washer method, which is more complex when revolving around the y-axis and integrating with respect to x. They might also forget the $2\pi x$ factor in the shell method or make errors in the u-substitution.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Differentiability Implies Continuity • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Consider the function $f(x)$ defined by $f(x) = \frac{x^2 - 9}{x - 3}$ for $xeq3$. Let $g(x)$ be a function such that $g(x) = \{f(x), \text{if } x < c; k, \text{if } x \geq c$. If $g(x)$ is continuous at $x = c$ and the limit of $g(x)$ as x approaches c from the left is 5, determine the value of k.

a. 3

b. **5 ✓**

c. 9

d. 12

For $g(x)$ to be continuous at $x = c$, the limit from the left must equal the limit from the right, and this must equal the function value at $x = c$. That is, $\lim_{x \to c^-} g(x) = \lim_{x \to c^+} g(x) = g(c)$. We are given that $\lim_{x \to c^-} g(x) = 5$. For $x < c$, $g(x) = f(x) = \frac{x^2 - 9}{x - 3}$. We can simplify this expression for $xeq3$: $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. Therefore, $\lim_{x \to c^-} g(x) = \lim_{x \to c^-} (x + 3)$. Since the limit from the left is given as 5, we have $c + 3 = 5$, which implies $c = 2$. Now consider the right side of the definition of $g(x)$. For $x \geq c$, $g(x) = k$. So, $\lim_{x \to c^+} g(x) = \lim_{x \to c^+} k = k$ and $g(c) = k$. For continuity, we must have $\lim_{x \to c^-} g(x) = \lim_{x \to c^+} g(x) = g(c)$. Substituting the values we found: $5 = k = k$. Thus, the value of k is 5.

39 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The function $f(x)$ is defined piecewise as: $f(x) = \{ax^2 + 1, , x < 2; 5x - b, , x \geq 2$ If $f(x)$ is continuous at $x = 2$, and the limit of $f(x)$ as $x \to 2$ is 9, find the values of a and b.

a. a = 2, b = 1

b. a = 1, b = 1

c. a = 2, b = -1

d. **a = 1, b = -1 ✓**

For $f(x)$ to be continuous at $x = 2$, the left-hand limit and the right-hand limit must be equal to the function value at $x = 2$. Given that the limit of $f(x)$ as $x \to 2$ is 9, we have: 1. Left-hand limit: $\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (ax^2 + 1) = a(2)^2 + 1 = 4a + 1$ 2. Right-hand limit: $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (5x - b) = 5(2) - b = 10 - b$ 3. Function value at $x = 2$: $f(2) = 5(2) - b = 10 - b$ Since the limit as $x \to 2$ is 9, we have: $4a + 1 = 9$ (from the left-hand limit) $10 - b = 9$ (from the right-hand limit and function value) Solving the first equation for a: $4a = 9 - 1$ $4a = 8$ $a = 2$ Solving the second equation for b: $10 - b = 9$ $-b = 9 - 10$ $-b = -1$ $b = 1$ Therefore, the values are $a = 2$ and $b = 1$.

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40 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

The slope of the tangent to $y = x^3$ at $x = 2$ is:

a. 6

b. 8

c. **12** ✓

d. 16

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $y' = 3x^2$; at $x = 2$: $3(4) = 12$ Therefore, the verified correct option is 12.

41 [Challenging] • Target: Difference Quotients & Limit Definitions • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find the derivative of $f(x) = 4e^x$.

a. e^x

b. $4e^x$ ✓

c. $-4e^x$

d. $4e^x + 4$

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $f'(x) = 4e^x$. Therefore, the verified correct option is $4e^x$.

42 [Challenging] • Target: Differentiability Implies Continuity • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find the derivative of $f(x) = 5e^x$.

a. $-5e^x$

b. $5e^x$ ✓

c. $5e^x + 5$

d. e^x

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $f'(x) = 5e^x$. Therefore, the verified correct option is $5e^x$.

43 [Challenging] • Target: Tangent and Normal Lines • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find the derivative of $f(x) = 3e^x$.

a. $-3e^x$

b. $3e^x + 3$

c. e^x

d. $3e^x$ ✓

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $f'(x) = 3e^x$. Therefore, the verified correct option is $3e^x$.

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44 [Challenging] • Target: Estimating Derivatives from Tables & Graphs • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find the derivative of $f(x) = 1\sec x$.

a. $\tan(x)\sec(x)$ ✓

b. $\sec(x)$

c. $\tan(x)\sec(x) + 1$

d. $-\tan(x)\sec(x)$

$f'(x) = \tan(x)\sec(x)$.

45 [Challenging] • Target: Difference Quotients & Limit Definitions • Cross-Lesson: 8.01 Area Between Curves

Find the derivative of $f(x) = 5\cos x$.

a. $5\cos(x)$

b. $5\sin(x)$

c. $-\sin(x)$

d. $-5\sin(x)$ ✓

$f'(x) = -5\sin(x)$.

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